DA4022 mid-project report · PENCIL extension · 15 October 2026

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How Much PENCIL Can You Afford?

Compute allocation for link prediction through candidate shortlisting and smaller sampled subgraphs

What we are testing

We rank candidate links with Adamic–Adar, run PENCIL only on the top-k candidates, and reduce the sampled subgraph budget s. **Sweeping (k, s) measures the accuracy retained at each GPU budget without retraining.**

1 AA ranking · CPU → **2 Top-k shortlist** · candidate count → **3 PENCIL** · context budget s

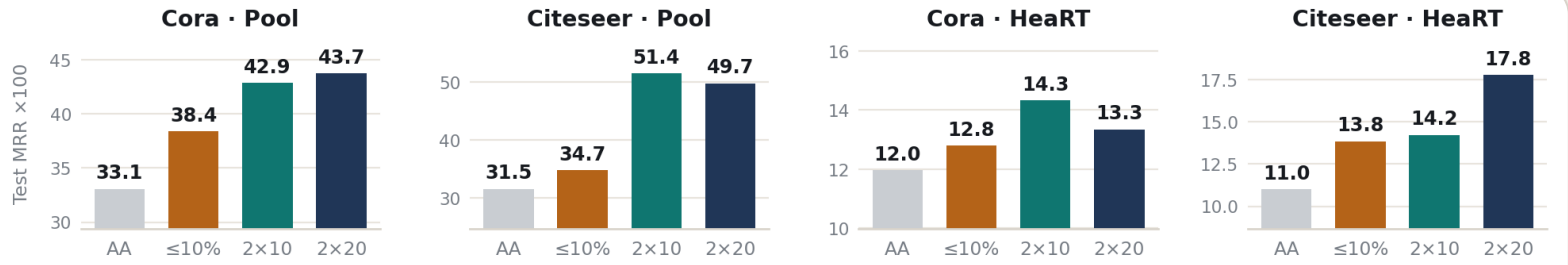
Experiment

Data	Cora and Citeseer
Evaluation	shared negative pool and HeaRT hard negatives
Selection	(k, s) chosen on validation data under fixed compute caps
Recorded	test MRR, measured T4 time, peak VRAM, three seeds

Reproduction check

44.00	**Cora test**	paper 42.23
50.46	**Citeseer test**	paper 47.51
112	**score dumps**	four regimes

Results



AA = CPU heuristic · ≤10% = validation-selected hybrid · 2×10 = half fanout · 2×20 = official full PENCIL

Test MRR at selected budgets

Policy	Cora Pool	Citeseer Pool	Cora HeaRT	Citeseer HeaRT
Chance	1.3	1.5	1.4	1.4
Adamic–Adar · 0%	33.1	31.5	12.0	11.0
Hybrid · ≤10%	**38.4 · 7%**	**34.7 · 9%**	**12.8 · 0.4%**	**13.8 · 8%**
PENCIL · s=2×10	**42.9 · 67%**	**51.4 · 63%**	**14.3 · 65%**	**14.2 · 69%**
PENCIL · s=2×20	43.7	49.7	13.3	17.8

What the sweep shows

01 Half fanout is usually enough

At s=2×10, PENCIL matches Cora Pool and beats full s=2×20 on Citeseer Pool and Cora HeaRT. Citeseer HeaRT is the only case that clearly needs full context.

02 The cheap hybrid helps everywhere

A validation-selected hybrid using at most 10% of full GPU compute improves over Adamic–Adar in all four test settings.

03 Compute matters in the difficult tail

Re-ranking the top 10% adds little. PENCIL is useful for links that graph heuristics cannot separate; 57.3% of true Cora test links have zero common neighbours.

04 The rule transfers across graphs

At a 25% cap, the rule selected on Cora reaches 44.9 MRR on Citeseer, compared with 37.9 from Citeseer's noisier local validation choice.

Operating points are selected on validation data; test data are used only for the reported results.